Resource Request algorithm.

```
#include<stdio.h>
#define MAX 10
int main(){
    int n,m;
    int allocation[MAX][MAX], max[MAX][MAX], need[MAX][M
int available[MAX];
    int request[MAX];
    int i,j,k,process;

    printf("Enter number of processes:");
    scanf("%d",&n);
    printf("Enter number of resource types:");
    scanf("%d",&m);
    printf("\nEnter allocation Matrix:\n");
     for(i=0; i<n;i++){
          for(j=0; j<n;j++){
               scanf("%d", &allocation[i][j]);
    printf("\nEnter Max Matrix:\n");
     for(i=0; i<n;i++)
        for(j=0;j<n;j++)
            scanf("%d", &max[i][j]);
    printf("\nEnter available resources:\n");
        for(j=0; j<m;j++)
            scanf("%d", &available[j]);

    for(i=0; i<n;i++)
        for(j=0; j<m;j++)
        need[i][j]=max[i][j]-allocation[i][j];
```

```
printf("\n Enter process number making request (0 to %d)
        :", n-1);
scanf("%d", &procnum);

printf(" Enter Request vector: \n");
for(j=0; j<m; j++)
    scanf("%d", &request[j]);

for(j=0; j<m; j++) {
    if(request[j] > need[procnum][j]) {
        printf("\n Error: Request exceeds process need!");
        return 0;
    }
}
for(j=0; j<m; j++) {
    if(request[j] > available[j]){
        printf(" Resource not available. Please wait.\n");
        return 0;
    }
}
for(j=0; j<m; j++){
    available[j] -= request[j];
    allocation[procnum][j] += request[j];
    need[procnum][j] -= request[j];
}
int work[MAX], finish[MAX]={0}, safeSeq[MAX];
int count = 0;

for(j=0; i<m; j++)
    work[j] = available[j];
```

```
while(count<n){
  int found=0;

  for(i=0;i<n;i++){
    if(finish[i]==0){
      int canExecute=1;

      for(j=0;j<m;j++){
        if(need[i][j]>work[j]){
          canExecute=0;
          break;
        }
      }
      if(canExecute){
        for(k=0;k<m;k++){
          work[k]+=allocation[i][k];
        safeSeq[count++]=i;
        finish[i]=1;
        found=1;
        }
      }
    }
    if(!found) break;
  }
  if(count==n){
    printf("\nRequest can be granted.\nSafe Sequence:");
      for(i=0;i<n;i++)
        printf("%d",safeSeq[i]);
      printf("\n");
  }else{
    printf("\nRequest cannot be granted.Rolling back\n");
```

```
for(j=0; j<m; j++){
    available[j] += request[j];
    allocation[process][j] -= request[j];
    need[process][j] += request[j];
  }
}
return 0;
}
```

output:

```
Enter number of processes: 5
Enter number of resource types: 3

Enter allocation matrix:
0 1 0
2 0 0
3 0 2
2 1 1
0 0 2

Enter Max Matrix:
7 5 3
3 2 2
9 0 2
2 2 2
4 3 3

Enter Available Resources:
3 3 2
```

Enter process number making request (0 to 4): 1
Enter request vector:
1 0 2

Request can be granted.
Safe sequence: P1 → P3 → P4 → P0 → P2.

Process	Allocation	Max	Available	Need
	A B C	A B C	A B C	A B C
P0	0 1 0	7 5 3	3 3 2	7 4 3
P1	2 0 0	3 2 2	2 3 0	1 2 2
P2	3 0 2	9 0 2	0 2 0	6 0 0
P3	2 1 1	2 2 2	3 0 1	0 1 1
P4	0 0 2	4 3 3	2 1 1	4 3 1